t3-w04 — Three Dimensional Shapes — Three dimensional figures: properties of cubes, cuboids, pyramids and cones

Run `diag-v3` · conditions C0 and C1 · the pack shown is draw 3 of each.

C0 `The curriculum objective alone. No lesson plan, no note.`

Input — what the model was sent

Contract `ComposeWriteObjectiveSection`, declaring `artefact_spec, lesson_context, unit_context`.

Lesson context — shared by every section

`subject`

Mathematics

`grade_level`

JSS1

`topic`

Three Dimensional Shapes

`subtopic`

Three dimensional figures: properties of cubes, cuboids, pyramids and cones

`term`

Third Term

`week`

4

`lesson_objectives`

• Identify the properties of cube and cuboids
• Identify the properties of pyramids and cones

Objective 1 — Identify the properties of cube and cuboids

`objective_text`

Identify the properties of cube and cuboids

Withheld from this payload: `step_title, step_content, instructional_type, estimated_minutes, current_step, instructional_events, student_tasks, activities, assessment_points, source_evidence, source_figures, objective_learner_state`.

Objective 2 — Identify the properties of pyramids and cones

`objective_text`

Identify the properties of pyramids and cones

Withheld from this payload: `step_title, step_content, instructional_type, estimated_minutes, current_step, instructional_events, student_tasks, activities, assessment_points, source_evidence, source_figures, objective_learner_state`.

Output — the pack it produced (draw 3)

Objective 1

Properties of Cubes and Cuboids

EXPLANATION

A **cube** is a three-dimensional shape with the following properties:
- 6 faces, and every face is a square.
- 12 edges, all of the same length.
- 8 vertices (corners).
- Opposite faces are parallel and congruent.

A **cuboid** (also called a rectangular prism) shares the same overall shape as a cube but its faces are rectangles. Its properties are:
- 6 faces, each a rectangle.
- Opposite faces are equal in size and parallel.
- 12 edges, grouped into three sets of four edges; each set has the same length (length, width, height).
- 8 vertices.
- If any two of the three dimensions are equal, the corresponding pair of faces become squares, but a cuboid is not required to have any square faces.

These properties let you identify whether a solid you see is a cube or a cuboid by looking at the shape of its faces and comparing edge lengths.

WORKED_EXAMPLE

Worked Example

Imagine you are given a solid that has 6 faces. Four of the faces are rectangles measuring 5 cm × 3 cm, and the other two faces are squares measuring 5 cm × 5 cm.

1. Count the faces: there are 6 – this matches both a cube and a cuboid.
2. Look at the shape of the faces:
   - Because some faces are rectangles and not all are squares, the solid cannot be a cube.
3. Check the edges:
   - The edges that belong to the 5 cm × 3 cm faces are of two lengths: 5 cm and 3 cm.
   - The edges that belong to the square faces are both 5 cm.
   - Thus we have three distinct edge lengths (5 cm, 3 cm, and 5 cm again), grouped into three sets of four equal edges.
4. Conclusion: the solid is a **cuboid**. Its properties are:
   - 6 rectangular faces (two happen to be squares because the length and width are equal).
   - Opposite faces are equal.
   - 12 edges in three groups of four equal lengths (5 cm, 3 cm, and 5 cm).
   - 8 vertices.

This process shows how to use the listed properties to decide whether a shape is a cube or a cuboid.

1. A solid has 6 faces, all of which are squares, and 12 edges of equal length. What is the solid called? List its key properties.
answer: Cube – 6 square faces, 12 equal edges, 8 vertices, opposite faces are parallel and congruent.

2. A solid has 6 rectangular faces, opposite faces are equal, and its 12 edges come in three groups of four equal lengths. What is the solid called? List its key properties.
answer: Cuboid – 6 rectangular faces, opposite faces equal, 12 edges in three sets of four equal lengths, 8 vertices.

3. How many vertices does a cube have? How many edges does a cuboid have?
answer: Cube: 8 vertices. Cuboid: 12 edges.

4. A cuboid has dimensions 3 cm × 4 cm × 5 cm. How many of its faces are squares?
answer: 0 square faces.

Properties of Pyramids and Cones

EXPLANATION

A pyramid is a solid that has a polygonal base and triangular faces that meet at a single point called the apex. The type of pyramid is named after its base: e.g., a square pyramid has a square base. For any pyramid:

- **Faces**: the base plus one triangular face for each side of the base. So a pyramid with an *n*-sided base has *n* + 1 faces.
- **Edges**: the base has *n* edges, and each vertex of the base is joined to the apex, giving another *n* edges. Total edges = 2*n*.
- **Vertices**: the *n* vertices of the base plus the apex, so *n* + 1 vertices.

A cone is a solid with a circular base and a single curved surface that narrows to a point called the apex. For a right circular cone:

- **Faces**: one circular face (the base) and one curved surface, so 2 faces.
- **Edges**: the circular base has one edge (the circle) and the curved surface meets the base along that edge, so 1 edge.
- **Vertices**: only the apex, so 1 vertex.

(If the cone is oblique, the same counts apply.)

WORKED_EXAMPLE

Example 1: Square Pyramid
- Base: square (4 sides)
- Number of faces: base + 4 triangular faces = 5 faces
- Number of edges: 4 edges of the base + 4 edges from base vertices to the apex = 8 edges
- Number of vertices: 4 base vertices + 1 apex = 5 vertices

Example 2: Right Circular Cone
- Base: circle
- Number of faces: 2 (one circular base, one curved surface)
- Number of edges: 1 (the circular edge where the base meets the surface)
- Number of vertices: 1 (the apex).

1. A triangular pyramid (tetrahedron) has a triangular base. How many faces, edges, and vertices does it have?
answer: Faces: 4, Edges: 6, Vertices: 4

2. A pentagonal pyramid has a pentagon as its base. List its number of faces, edges, and vertices.
answer: Faces: 6, Edges: 10, Vertices: 6

3. For a right circular cone, state the number of faces, edges, and vertices.
answer: Faces: 2, Edges: 1, Vertices: 1

4. If a pyramid has a hexagonal base, how many edges does it have?
answer: 12

C1 `The objective plus the lesson plan's design: position, method labels, materials, entry behaviour and assessment.`

Input — what the model was sent

Contract `ComposeWriteObjectiveSection`, declaring `artefact_spec, lesson_context, unit_context`.

Lesson context — shared by every section

`subject`

Mathematics

`grade_level`

JSS1

`topic`

Three Dimensional Shapes

`subtopic`

Three dimensional figures: properties of cubes, cuboids, pyramids and cones

`term`

Third Term

`week`

4

`lesson_objectives`

• Identify the properties of cube and cuboids
• Identify the properties of pyramids and cones

`entry_behaviour`

• Students have been taught the differences between cube and cuboid.

`instructional_materials`

• Shapes of cubes, cuboid, Pyramid, cone, cylinder and spheres.

Objective 1 — Identify the properties of cube and cuboids

`objective_text`

Identify the properties of cube and cuboids

`current_step`

1

`instructional_events`

• present_content

`assessment_points`

• How many vertices, has a cuboid?
• How many edges has a cube?
• Draw a cube and cuboid then label.

Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `student_tasks`, `activities`, `source_evidence`, `source_figures`, `objective_learner_state`.

Objective 2 — Identify the properties of pyramids and cones

`objective_text`

Identify the properties of pyramids and cones

`current_step`

2

`instructional_events`

• present_content

`assessment_points`

• Draw an hexagonal base pyramid and list the properties.
• How many vertices has a tetrahedron.

Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `student_tasks`, `activities`, `source_evidence`, `source_figures`, `objective_learner_state`.

Output — the pack it produced (draw 3)

Objective 1

Properties of Cubes and Cuboids

> EXPLANATION
> A **cube** is a special type of cuboid where all six faces are squares. This means:
> - **Faces**: 6 (all squares)
> - **Edges**: 12, and every edge has the same length.
> - **Vertices**: 8, where three edges meet at each corner.
>
> A **cuboid** (also called a rectangular prism) has six rectangular faces. Its properties are:
> - **Faces**: 6 (pairs of opposite faces are equal rectangles).
> - **Edges**: 12, but opposite edges are equal in length; not all edges are the same.
> - **Vertices**: 8, just like a cube.
>
> Both shapes are three-dimensional and have the same number of faces, edges, and vertices; the difference lies in the shape of the faces and the equality of the edges.

> WORKED_EXAMPLE
> **Worked Example – Counting the parts of a cube**
>
> 1. **Vertices** – Look at the corners of the cube. There are 4 corners on the top face and 4 on the bottom face, giving $4 + 4 = 8$ vertices.
> 2. **Edges** – Each face of the cube has 4 edges. However, each edge belongs to two faces, so counting each face's edges would double-count. A quicker way: a cube has 12 edges (4 along the top, 4 along the bottom, and 4 vertical edges connecting the top and bottom).
> 3. **Faces** – A cube has 6 faces: top, bottom, front, back, left, and right.
>
> Thus, a cube has **8 vertices, 12 edges, and 6 faces**.

1. How many vertices does a cuboid have?
answer: 8

2. How many edges does a cube have?
answer: 12

3. How many faces does a cuboid have?
answer: 6

4. Draw a cuboid and label its vertices, edges, and faces. How many of each should you label?
answer: Your drawing should show 8 vertices, 12 edges, and 6 faces, each correctly labeled.

Properties of Pyramids and Cones

EXPLANATION

A **pyramid** is a solid that has a polygonal base and triangular faces that meet at a single point called the apex. The number of triangular faces, edges and vertices depends on the shape of the base:

- **Base** – any polygon (triangle, square, pentagon, hexagon, …).
- **Faces** – one triangular face for each side of the base, plus the base itself.
- **Edges** – the edges of the base plus one edge from each base vertex to the apex.
- **Vertices** – the vertices of the base plus the apex.

For example, a pyramid with a hexagonal base has:
- 1 hexagonal base
- 6 triangular faces
- 12 edges (6 around the base, 6 from base vertices to the apex)
- 7 vertices (6 on the base, 1 apex)

A **tetrahedron** is a special pyramid whose base is a triangle. It therefore has 4 triangular faces, 6 edges and 4 vertices.

A **cone** is a solid with a circular base and a single curved surface that meets at a point called the vertex. Its main properties are:
- **Base** – a circle.
- **Vertex** – the point where all the straight lines from the edge of the base meet.
- **Height** – the perpendicular distance from the vertex to the centre of the base.
- **Slant height** – the distance from the vertex to any point on the edge of the base (useful for surface-area calculations).
- **Lateral surface** – the curved surface connecting the base to the vertex.

Both pyramids and cones are characterised by having one flat base and a single point where all side edges (or the curved surface) converge.

1. Draw a pyramid with a hexagonal base and list all of its properties (type of base, number of faces, edges and vertices).
answer: Base: hexagon; Faces: 7 (1 hexagonal base + 6 triangular faces); Edges: 12 (6 around the base + 6 from base vertices to the apex); Vertices: 7 (6 on the base + 1 apex).

2. How many vertices does a tetrahedron have?
answer: 4 vertices.

3. List the main properties of a right circular cone.
answer: Base: circle; Vertex: single point opposite the base; Height: perpendicular distance from vertex to centre of base; Slant height: distance from vertex to any point on the circular edge; Lateral surface: curved surface connecting base to vertex.

Result — per rubric dimension

Scores are 1–5, each the mean of 2 judge pass(es) on the same packet. The first three columns are the packs shown above (draw 3); the last is the lesson's mean over its 3 draws per condition — the study's number.

Rubric	C0	C1	C1 – C0, shown packs	C1 – C0, mean of 3 draws
Objective and curriculum alignment	5.00	5.00	+0.00	+0.00
Taught-lesson alignment	3.00	4.00	**+1.00**	**+0.67**
Scope and conceptual correctness	4.00	5.00	**+1.00**	**+0.83**
Student-facing clarity and usefulness	4.00	5.00	**+1.00**	**+0.50**